Demo Client · DEMO-0009
11 November 2026 · v1.0

Field	Value
Document	Methodology of the fraud and anomaly review
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1. Purpose

The review measures what the client's current fraud controls really catch, what they miss, and what the team should do with the capacity it has. It produces a score, a rule pack and a queue of cases. Alerts are review leads. The review does not establish that any person or business did anything wrong; the compliance team takes every reporting decision.

2. Inputs

Data set	Size	Fingerprint (SHA-256)
transactions	100,939 rows	`59c79c357ce5872c`
accounts	3,000 rows	`9e6df748cb753a9c`
merchants	420 rows	`43b8424681529e28`
fx_rates	734 rows	`0d53d2d5aba20d07`
alerts	2,592 rows	`a5d41ff79cd234e5`
cases	546 rows	`76d5257c1660ec05`

Each input is fingerprinted with SHA-256 on ingest and recorded in the run audit trail.

3. Cleaning

Exact duplicate rows are removed and counted. Timestamps are read in both formats; currency spellings are harmonised. A day on which a source system sent minor units (its median amount is many times its usual level) is divided by 100 and logged. Amounts are converted with the rate of the day; a missing rate is filled forward and flagged. A transaction is **excluded**, and listed with its reason, when it cannot be analysed honestly: unknown account, unreadable time or amount, dated after the period, unknown currency. Refunds are kept apart and are not scored. In the cases, outcome spellings are harmonised, repeated entries removed, a changed outcome resolved in favour of the latest entry, and cases pointing to no transaction listed.

4. Features with no look-ahead

Debits are scored; credits inform the features of the debits that follow them. Every feature of a transaction uses only what was known just before it: windows are closed on the left, "first time" flags compare with the past only, and the fraud history of a counterparty counts only cases already reported at that moment. A test recomputes the features of a transaction after deleting everything later and requires the same values. Groups: amount against the account's history; velocity (transactions and value in 1 and 24 hours); time since the last transaction and dormancy; new device, payee and country; accounts seen on the device; foreign and IP mismatch; impossible travel; night; round amounts; amounts just under the reporting threshold and their repetition; money received in the last 24 hours and money forwarded soon after it arrived; age of the account and of the merchant; small-amount bursts and accounts at a merchant in the last hour; accounts paying a payee in 7 days.

5. Leakage screen

Each numeric field is tested alone against the confirmed cases. A field that almost separates them by itself (AUC above the agreed limit) was probably filled in after the event. Such a field is reported and never used. Being strong is not enough to prove leakage: the analyst asks the data owner when the field is filled in.

6. Detectors and the combined score

Rules: a pack of transparent rules, each a list of conditions on the features, delivered in YAML and SQL. **Anomaly model:** an isolation forest fitted on training transactions with no confirmed fraud; it needs no labels, so it can notice what nobody has confirmed. **Supervised model:** gradient boosting fitted on confirmed cases in the training window, with balanced class weights; unconfirmed transactions count as negative, which is a known weakness. Scores are turned into percentiles p (supervised, against the validation window) and q (anomaly, against the training window), and combined as 1 - (1 - p)^w x (1 - q)^(1 - w), so that one very strong signal is enough. A rule of high severity sets a floor. The score shown is this value on a scale of 0 to 999.

7. Time windows and capacity

The first days only build up the history of each account (burn-in). The mature period ends a stated number of days before the extract, because cases are reported late; the rest is scored but not measured. The mature period is split in time order into training, validation and test. The threshold is the score at which the validation window gives the number of alerts the team can review, agreed at kickoff as analysts x alerts per analyst per day. The test window is seen once, at the end.

8. Measurement

On the test window, for the current rules, the rule pack, each model at the alert budget and the combined score: alerts a day, confirmed cases found, precision, recall by count and by value, each with a 95% bootstrap interval resampling transactions. **Confirmed cases are late, partial and biased**: precision is a lower bound because most alerts have never been reviewed, and the comparison favours the current rules because part of the cases exist only because those rules fired. The sweep shows what each extra alert buys.

9. Current rules

For each rule: alerts, alerts a day, reviewed, open, open beyond the SLA, confirmed, precision among the reviewed and value found. A rule with no confirmed fraud is a candidate for retirement after a documented review.

10. Fairness

On transactions with no confirmed fraud, the rate of alerts by segment and by home country, against the overall rate. A false alert has a cost for the customer, and a gap has to be understood. The dimensions and any protected characteristics are agreed with compliance; a gap is not automatically wrong, and compliance decides whether it is justified.

11. Drift

Population Stability Index of the score and of each feature, from the training to the test window.

12. Rings and merchants

Accounts are joined when they use the same device (union-find), leaving out devices on very many accounts, which are reported on their own as a compromised-device pattern; groups that send most of their transfers to the same beneficiary are joined. A group of at least three accounts is a **suspected ring** when transfers out are a high share of the credits in and the median time from credit to transfer is short. Merchants are screened on off-hours trading, round amounts, small-amount bursts, the first-digit test (mean absolute deviation from Benford's law) and age; a merchant is suspected when several signals agree or a small-amount burst is seen. None of these is proof.

13. Cases

Alerts since the start of the test window, plus the transfers of suspected rings, are grouped by ring, then merchant, then account. Priority: P1 for a ring, a merchant or an exposure above the limit; P2 for a smaller exposure or a very high score; P3 for the rest. Reasons in plain language come from the rules that fired and from the features furthest from normal, weighted by the importance the model gives them.

14. Economics

Prevented = confirmed value found x share that can still be stopped. Cost = alerts x cost of a review + false alerts x friction. Scaled to a year from the test window. The team-size table compares analysts at their annual cost. Confirmed fraud is a lower bound, so the benefit is conservative.

15. Assessment

Twenty-five tests in six areas (data, detection, current controls, fairness, stability, exposure) give PASS, WARN or FAIL against thresholds agreed at kickoff; the overall view is Effective (all pass), Effective with actions (some warn) or Material gaps (some fail), with the action each non-passing test implies.

16. Commentary

Sentences are assembled by rules from the tables. The analyst reads the draft and may add notes.

17. Reproducibility

Inputs, configuration and code version are recorded in `audit.json`. Models and the bootstrap use fixed seeds. Every table in the deliverables is a CSV in the run folder.

18. Limitations

The limits of this work are set out in the report. No deviations from the standard methodology.